

18 22 24 25 27 28

$$\boxed{5.18} \quad \tilde{S} = \tilde{U} \tilde{I}^* = (15\sqrt{2} + j 15\sqrt{2})(10 \cos 15^\circ - j 10 \sin 15^\circ)$$

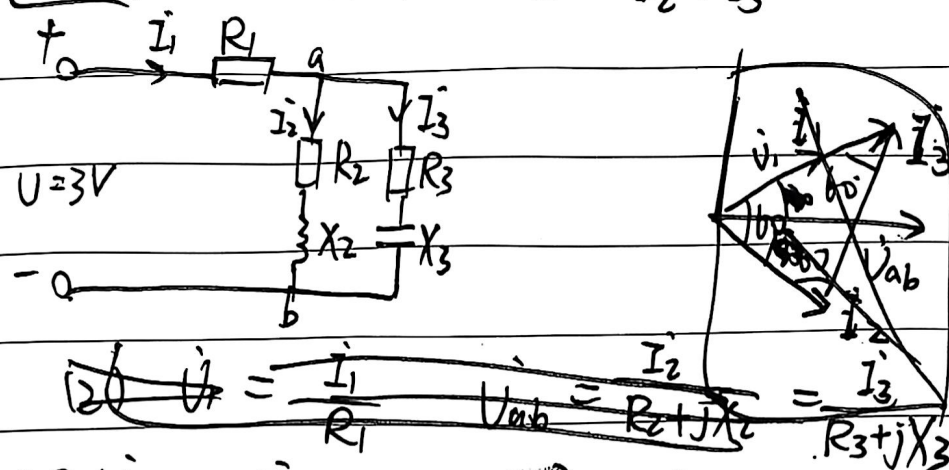
$$= 150\sqrt{2} + j 150 \text{ (J)}$$

$$\therefore P = \cancel{150} 259.8 \text{ J}$$

$$\therefore Q = 150 \text{ J}$$

$$\therefore S = \sqrt{P^2 + Q^2} = 300 \text{ J}$$

$$\boxed{5.22} \quad I_1 = 1 \text{ A} \quad \tilde{I}_1 = \tilde{I}_2 + \tilde{I}_3$$



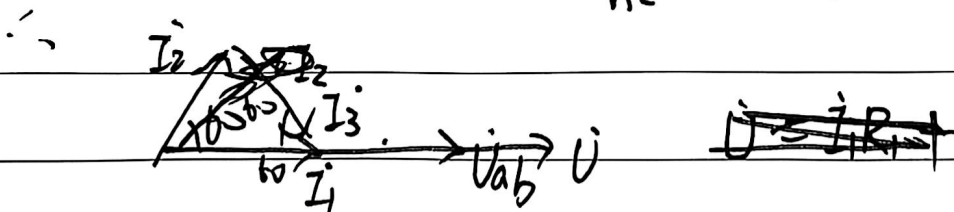
"X3" 并不包括方向
应指明 $-jX_3$

$$\cancel{(2) \quad \tilde{U} = \frac{\tilde{I}_1}{R_1} \quad V_{ab} = \frac{\tilde{I}_2}{R_2 + jX_2} = \frac{\tilde{I}_3}{R_3 + jX_3}}$$

$$(1) \quad V_{ab} = \tilde{I}_2 (R_2 + jX_2) = \tilde{I}_3 (R_3 + jX_3)$$

$$V_{ab} = \tilde{I}_2 \sqrt{R_2^2 + X_2^2} = \tilde{I}_3 \sqrt{R_3^2 + X_3^2} \quad \therefore X_2 = X_3$$

$$\therefore \tilde{I}_2 + \tilde{I}_3 = \tilde{U}_{ab} \left(\frac{R_2 - jX_2}{R_2^2 + X_2^2} + \frac{R_3 - jX_3}{R_3^2 + X_3^2} \right) = \tilde{I}_1 \angle 0 = \tilde{I}_1$$

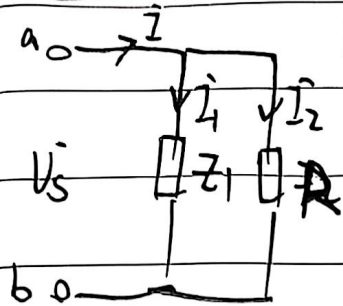


$$(2) \quad I_1 = 1 \text{ A} = I_2 = I_3$$

$$\therefore \begin{cases} \sqrt{R_2^2 + X_2^2} = \frac{V_{ab}}{I_2} = \frac{U - I_1 R_1}{I_2} = 3 - R_1 = \sqrt{R_1^2 + X_2^2} \\ X_2 = \sqrt{3} R_1 \end{cases} \Rightarrow \begin{cases} R_1 = 1 \Omega \\ X_2 = \sqrt{3} \Omega \\ X_3 = \sqrt{3} \Omega \end{cases}$$

5.24

$$\omega = 100\pi, U_s = 100V, I_1 = 10A, \cos\varphi_1 = 0.5, R = 20\Omega$$



(1) 设 $\dot{U}_s = 100\angle 0$

$$\cos\varphi_1 = 0.5, \varphi_1 = \pm\frac{\pi}{3}, \text{感性} \Rightarrow \varphi_1 = \frac{\pi}{3}$$

$$\therefore \dot{I}_1 = 10\angle -\frac{\pi}{3}$$

$$\therefore Z_1 = \frac{\dot{U}_s}{\dot{I}_1} = 10\angle \frac{\pi}{3} = 5 + 5\sqrt{3}j$$

$$\therefore P = I_1^2 \operatorname{Re}(Z_1) + \frac{U_s^2}{R} = 1000W$$

$$\therefore \dot{I}_2 = \frac{\dot{U}_s}{R} = 5\angle 0$$

$$\therefore \dot{I} = \dot{I}_1 + \dot{I}_2 = 10\angle -\frac{\pi}{3} + 5\angle 0 = 10 - 5\sqrt{3}j$$

$$I = 5\sqrt{7}A = 13.23A$$

$$\therefore \cos\varphi = \frac{P}{UI} = 0.756$$

(2) U_s 不变, I_1, I_2 不变

$$\dot{I}_3 = \frac{\dot{U}_s}{X_c} = j\omega C \cdot \dot{U}_s$$

$$\therefore \dot{I} = \dot{I}_1 + \dot{I}_2 + \dot{I}_3 = 10 + j(100\omega C - 5\sqrt{3})$$

$$\therefore I = \sqrt{100 + (10000\pi C - 5\sqrt{3})^2} = 11$$

$$\Rightarrow C = 4.215 \times 10^{-4}F \text{ or } 1.298 \times 10^{-4}F$$

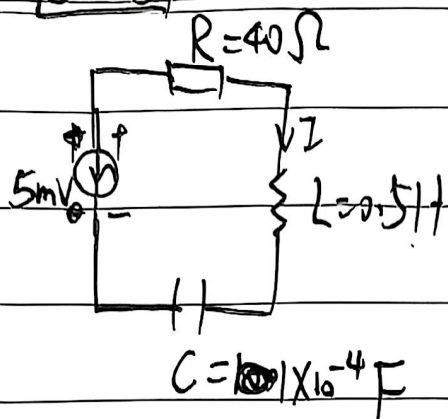
$$C_{\min} = 1.298 \times 10^{-4}F$$

~~$$\cos\varphi = \frac{P}{UI}$$~~

$$\cos\varphi = \frac{P}{UI} = \frac{10}{11} = 0.91$$

5.25

方向?



$$\omega_0 L = \frac{1}{\omega_0 C} \Rightarrow \omega_0 = \sqrt{\frac{1}{CL}} = 141.4$$

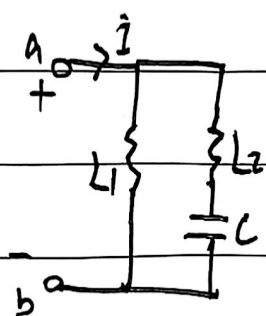
$$f = \frac{\omega}{2\pi} = 22.5 \text{ Hz}$$

$$I = \frac{U}{R} = 1.25 \times 10^{-4} \text{ A}$$

$$\therefore U_L = U_C = I \cdot \omega_0 L = 1.5625 \times 10^{-3} \text{ V}$$

$$\therefore Q = \frac{U_C}{U} = 1.76$$

5.27



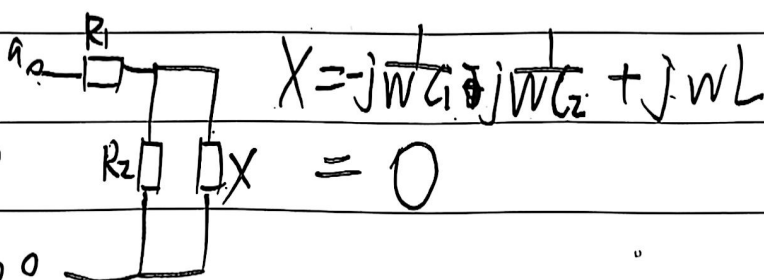
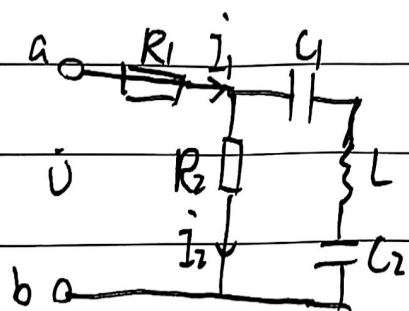
$$Y = \frac{1}{j\omega_0 L_1} + \frac{1}{j\omega_0 L_2 - j\frac{1}{\omega_0 C}} = -j \left[\frac{\omega_0^2 C (L_1 + L_2) - 1}{\omega_0 L_1 (\omega_0^2 L_2 C - 1)} \right]$$

$$\omega_0 = \sqrt{\frac{1}{C(L_1 + L_2)}} = 1 \times 10^3 \text{ rad/s}$$

$$\omega_0 = \sqrt{\frac{1}{L_2 C}} = \sqrt{\frac{5}{3}} \times 10^3 \text{ rad/s} = 1.29 \times 10^3 \text{ rad/s}$$

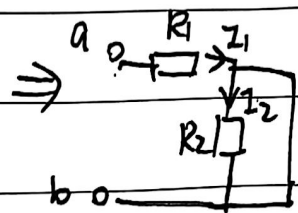
5.28

$$\dot{U} = 220 \angle 20^\circ \quad R_1 = R_2 = 10 \Omega \quad C_1 = 2 \times 10^{-5} \text{ F} \quad C_2 = 8 \times 10^{-5} \text{ F}$$



$$X = -j\omega C_1 + j\omega C_2 + j\omega L$$

$$= 0$$



$\therefore R_2$ 被短路. $I_2 = 0$

$$I_1 = \frac{\dot{U}}{R_1} = 2 \angle 20^\circ \text{ A}$$

$$\therefore X_{\text{输入}} = R_1 = 10 \Omega$$